⚡ Charges & Static: Reading

Chemistry · Unit 1A · Level 3 (closer to the class work) · goes with L1.1-L1.6 (Initial Model, Smaller Scales, Sparks, Particle Model of Static, Cause of Static)

Name: _____

What Causes Lightning?

Lightning is a giant electric spark. It can jump from a cloud to the ground or between clouds. It heats the air to about 30,000 °C, about five times hotter than the surface of the sun. The hot air expands so fast that it makes a boom: thunder.

Scientists noticed that clouds that make a lot of rain also make a lot of lightning. This is a correlation: the two things happen together. A correlation does not prove that one thing causes the other. To find the cause, we need a model at a smaller scale.

All matter is made of atoms. In the center of each atom is the nucleus. It holds protons, which have a positive (+) charge, and neutrons, which have no charge. Around the nucleus are electrons, which have a negative (−) charge. A neutral atom has the same number of protons and electrons.

Protons are locked in the nucleus, but electrons can move from one object to another. When you rub a balloon on your hair, electrons move from your hair to the balloon. The balloon now has extra electrons, so it is negative. Your hair is missing electrons, so it is positive. This build-up of charge is called static charge.

Inside a storm cloud, ice crystals and soft hail (graupel) bump into each other as air moves up and down. Each bump moves electrons, like rubbing a balloon. The bottom of the cloud becomes negative and the top becomes positive. When enough charge builds up, electrons jump through the air. That jump is lightning.

Check your reading

1. What is the difference between a correlation and a cause?

2. Why do electrons, and not protons, move when you rub a balloon?

3. How is a storm cloud like a rubbed balloon?